

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		(Birth rate + immigration) = (death rate + emigration) (1)		1		1		
	(b)	(i)		159/ 160 (2) Award 1 mark for (19 ×143)/17 159.8		2		2	2	2
		(ii)		Removing {part of digit/ toe} might affect {ability to survive/ increase chance of recapture/ owtte} (1)			1	1		1
	(c)			Shape – rods (1) reject cylinders Colour – pink/red (1)		2		2		
	(d)	(i)		190 or 180 or 170 = 3 marks Award 2 marks for 186.9565217 / any number that gives a 2 sf of 170 180 or 190 Award 1 mark for Sight of correct calculation for gradient = 4300/23/ or any gradient correctly indicated on the graph		3		3	3	3
		(ii)		- Because Pond 1 has more tadpoles (than Pond 2) (1) - the {rate of decline/ population decrease} was faster in high density pools /use of data (1) - Because of {competition for food/ predation/ disease/ accumulation of (toxic) waste} (1)			3	3		

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	(e)	(i)		Selection increases {rate of development/ body length/ rate of growth} (1) Any one (x1) from {Rate of development/ growth} is faster if density is low/OR (1) The less tadpoles in an area the longer they get (1) Less time needed for development if density is low/ OR (1) The lower the density the higher the rate of development (1) Longer body length in the low density tank/ owtte (1)			2	2		2
		(ii)		• Removing tadpoles would {reduce/ change} the {density/ population}/ owtte (1) • Remove same number of tadpoles of random size from {the high density tank/ tank 2}/ Replace the two removed with two of {mean/ random} size (1)			2	2		2
				Question 4 total	0	8	8	16	5	10